

Show all of your work in the space provided to you. Full credit will not be awarded for correct answers unless sufficient supporting work is presented.

(16 pts) Let $A = \begin{bmatrix} 1 & -2 & -4 & 0 & -2 \\ 0 & 2 & 6 & 1 & 6 \\ 1 & 0 & 2 & 1 & 4 \\ 3 & 1 & 9 & 0 & 8 \end{bmatrix}$ and $\text{RREF}(A) = \begin{bmatrix} 1 & 0 & 2 & 0 & 2 \\ 0 & 1 & 3 & 0 & 2 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$.

a) What are the rank and the nullity of A

rank =

nullity =

b) Find a basis for the column space of A.

c) Find a basis for the null space of A.

d) Is $\text{Col } A$ equal to $\mathbf{R}^4$? Give a reason for your answer.

2. (10 pts) Let $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$ with $\det A = -2$. Find the following determinants.

a. $\det A^{-1} =$

b. $\det A^T =$

c. $\det A^3 =$

d. $\det (3A) =$

e. $\det \begin{bmatrix} a & 5b & c \\ d & 5e & f \\ g & 5h & i \end{bmatrix} =$

3. (8 pts) Let A be any $n \times n$ invertible matrix. Use properties of determinants and inverses to prove that $A^T A$ is also invertible.

4. (6 pts) Let B be any 7×3 matrix with at least one nonzero entry.

a) What is the largest possible dimension of $\text{Col } B$?

b) What is the largest possible dimension of $\text{Nul } B$?

5. (10 pts) Let $\mathbf{B} = \{\mathbf{V}_1, \mathbf{V}_2, \mathbf{V}_3, \mathbf{V}_4\}$ be a set of vectors in $\mathbf{R}^4$.

(4 pts) What two properties must $\mathbf{B}$ satisfy to be a basis for $\mathbf{R}^4$?

(3 pts) Explain carefully why any set of fewer than 4 vectors cannot be a basis for $\mathbf{R}^4$

(3 pts) Explain carefully why any set of more than 4 vectors cannot be a basis for $\mathbf{R}^4$

6. (10 pts) Let $\mathbf{B} = \{\mathbf{b}_1, \mathbf{b}_2\}$ with $\mathbf{b}_1 = \begin{bmatrix} -3 \\ 2 \\ -4 \end{bmatrix}$ and $\mathbf{b}_2 = \begin{bmatrix} 7 \\ -3 \\ 5 \end{bmatrix}$ be a basis for a subspace H in

$\mathbf{R}^3$ and let $\mathbf{x} = \begin{bmatrix} 5 \\ 0 \\ -2 \end{bmatrix}$ Find the coordinate vector for $\mathbf{x}$ relative to the basis $\mathbf{B}$.

7. (10 pts) Let $A = \begin{bmatrix} 1 & -1 & 2 \\ 1 & -3 & 1 \\ 3 & 7 & 5 \end{bmatrix}$. Find an LU factorization for A and explain how the factorization is used to solve the equation $A\mathbf{x} = \mathbf{b}$.

8. (12 pts) Let T be any linear transformation from vector space V to vector space W . Recall that the kernel of $T = \{\mathbf{x} \text{ in } V : T(\mathbf{x}) = \mathbf{0}\}$. Prove that the kernel of T is a subspace of V .

9. (12 pts) Let T be the function from $\mathbf{P}^3$ to $\mathbf{P}^2$ that maps any polynomial, f , in $\mathbf{P}^3$ to its derivative f' in $\mathbf{P}^2$.

(9 pts) Prove that T is a linear transformation. You may use any facts from calculus that you know.

(3 pts) Describe all the functions in $\mathbf{P}^3$ that are in the kernel of T .

10. (6 pts) **Let A be an $n \times n$ matrix.** Circle each letter beside any statement listed below that is equivalent to the statement “ A is an invertible matrix.”

- a) There is a pivot in every row of $\text{RREF}(A)$.
- b) There is a pivot in every column of $\text{RREF}(A)$.
- c) The matrix equation $A\mathbf{x} = \mathbf{b}$ has a unique solution for every vector $\mathbf{b}$ in $\mathbf{R}^n$.
- d) A is row equivalent to the $n \times n$ identity matrix.
- e) The span of the columns of A is equal to $\mathbf{R}^n$.
- f) The determinant of A is not zero.